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# The half-normal specification for the two-tier stochastic frontier model

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***Summary:*** *In the context of the two-tier stochastic frontier model, we present a new closed-form stochastic specification, the half-normal one. Expressions to calculate average and conditional measures of inefficiency are provided, for regression specifications in levels and in logs. Some results related to model validation and maximum likelihood estimation are also presented.*

***Keywords****: Two-tier stochastic frontier analysis, half -normal distribution, maximum likelihood, informational efficiency.*

***JEL CODES****: C16, C21, D82, D83.*

## 1. Introduction and literature review

What distinguishes stochastic frontier (SF) analysis from an econometric point of view is the composite error specification where, alongside the usual zero-mean symmetric disturbance, a one-sided error term appears (positive or negative depending on the context), which is exactly the component that is used to measure the subject matter of SF analysis, the deviations from full efficiency assumed by economic theory.

Polachek and Yoon (1987) have introduced a three-component error structure with two one-sided terms, one positive and one negative, in order to capture imperfections on both sides of a transaction (the "Two-tier Stochastic Frontier Model", hereafter 2TSF model). They originally applied their model to the labor market in order to estimate the earnings equation, where the two one-sided error terms were interpreted as representing the quantitative impact of informational deficiencies of employers and of workers separately, leading the former to pay more than workers' reservation wages, and the latter to work for less than maximum wages acceptable by employers. Groot and Oosterbeek (1994), using an earnings-related data set from Netherlands, extended the model to be able to link informational inefficiencies to individual characteristics, by assuming that the moments of each one-side disturbance is a

function of the regressors or other variables (technically, this amounts to assume heteroskedastic one-sided errors). In the same spirit but using another route, Polachek and Yoon (1996) applied their original model to panel data in order to disentangle unobserved individual heterogeneity from the informational imperfections. They found that the panel data approach improved the quality of the results, and that the significance of informational inefficiencies, although it was reduced, persisted. Sharif and Dar (2007) applied the original 2TSF model in cross-sectional earnings data from Canada focusing on immigrants, while Murphy and Strobl (2008) applied the model as extended by Groot and Oosterbeek to data from Trinidad & Tobago. The results of these studies are mixed, regarding the sources of deviations from full-information wage. In some, "employers' ignorance" emerges as the main factor leading the wage above its perfect information level, in others, it is "employees' ignorance" that dominates and leads the wage below it. But in all cases, the quantitative significance of these informational inefficiencies is clear.

The 2TSF model has also seen applications in the Health Services market, where informational asymmetries and deficiencies are believed to abound. Gaynor and Polachek (1994) applied it to the physician services' market in an attempt to explain wide price variations, while Chawla (2002) used it on data from Egypt, and found that doctors were extracting a large surplus in their transactions with the patients. Tomini et al. (2012) used the model to measure the effects of incomplete information on the informal (under-the-table) payments made to physicians in Albania. Interestingly, they found that in this case the deficiencies were in favor of patients, against conventional wisdom. But this result accords with intuition, and strengthens the case in favor of the 2TSF model as a valid method to unearth existing phenomena: in the absence of a public market and price system, buyers-patients tend to exchange more information regarding informal costs, while on the other hand sellers possess *less* information for the same reasons, and also tend to avoid exchanging information because informal revenues are considered unethical and usually are illegal.

Recently, the application of the 2TSF model seems to be picking pace, and it starts to be used in innovative ways and sometimes in highly unexpected contexts, thus showing its flexibility and potential. Kumbhakar and Parmeter (2009) used the model to measure the effects of bargaining over wage in a framework of productivity uncertainty, while Kinukawa and Motohashi (2010) followed their lead and bargaining model but with a paper on alliances in the biotechnology market. Kumbhakar and Parmeter (2010) presented a 2TSF generalized hedonic price model with an application to the house market. Ferona and Tsionas (2012) applied the model to an auction setting, using data from timber auctions, in order to assess the extent of underbidding and overbidding behavior. But also Groot and van den Brink (2007) used it to measure "optimism" and "pessimism" in assessing quality of life, while Poggi (2010) re-applied the model to the labor market, but this time in order to measure job satisfaction. Researchers in China seem also to have taken an interest in the model: Lian & Chung (2008) and Yu and Liang (2012) applied it to investigate the effects of financing constraints and agency costs on investment behavior and on dividend policies respectively, of Chinese firms. Also, Zheng and Zhang (2012), and Huang (2013) used the model to separate the "premium" effect from the "underpricing" effect in IPOs of Chinese firms. In all cases the 2TSF model has delivered significant results, providing quantitative insights into the magnitude and effects of imbalances and conflicting forces of various kinds on both sides of economic transactions (or of psychological matters).

We aim at contributing to this momentum with the present paper, the remaining of which is structured as follows: in Sect. 2 we present a new stochastic specification for the composite error term in closed form, namely the half-normal specification. In Sect. 3 we present measures of the effects of inefficiencies, at various levels of aggregation, and for different regression specifications. In Sect. 4 we provide some discussion on model selection and also formulas to be used in the context of maximum likelihood estimation, completing the basic toolkit needed to implement the new specification. Section 5 concludes, with

suggestions for further research. For reasons of brevity, proofs of the various results are omitted[1].

## 2. The half-normal specification for the 2TSF model

Without any exception, all studies mentioned in the previous section have assumed the same distributional specification: a normal symmetric disturbance, and two exponential one-sided errors. Polachek and Yoon introduced this stochastic specification emphasizing tractability. Moreover, for the basic, one-tier stochastic frontier specification, experience had shown that, at least qualitatively, it mattered little whether the one-sided term was assumed to follow an exponential or a half-normal distribution. But this similarity does not fully extend to the case of the 2TSF model. Tsionas (2012) compared results from a 2TSF model under the normal-exponential error specification, against results under a normal-gamma specification (estimated by a Fourier Transform algorithm that approximates distributional specifications unavailable in closed form). He found that the exponential specification, which is nested by the gamma one, was rejected for at least one of two one-sided random variables, and also that there were significant differences in the estimates of the terms related to the error components and their implications for the economic phenomenon under study. Although this is just one instance of inadequacy of the available exponential specification, it is somehow impressive that it happened the first time it was subjected to such a trial. Therefore, providing an alternative specification appears worthwhile. We chose the (normal)-half normal specification since it is the prevailing one in the one-tier model.

We specify in abstract terms a single-equation cross-sectional regression model, linear in the parameters,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \ , \quad \boldsymbol{\varepsilon} = \mathbf{u} + \mathbf{v} - \mathbf{w} \qquad [1]$$

where $\mathbf{y}$ is a $n \times 1$ vector containing observations of the dependent variable, usually in levels or in logs, $\mathbf{X}$ is a $n \times K$ matrix of full column rank containing observations of the

[1] All proofs and calculations related to the results presented in the paper are available from the author upon request.

regressors (including a constant term), and $\boldsymbol{\varepsilon} = \mathbf{u} + \mathbf{v} - \mathbf{w}$ is the $n \times 1$ vector of the composite error term. The error term components are assumed independent from one another for each cross-section, and identically and independently distributed over cross-sections, with typical element for $i = 1, ..., n$

$$u_i \sim N\left(0, \sigma_u^2\right),\ v_i \sim HN\left(\sqrt{\frac{2}{\pi}}\sigma_1, \left(1-\frac{2}{\pi}\right)\sigma_1^2\right),\ w_i \sim HN\left(\sqrt{\frac{2}{\pi}}\sigma_2, \left(1-\frac{2}{\pi}\right)\sigma_2^2\right)$$

where $\sigma_1^2$, $\sigma_2^2$ are the variances of the symmetric zero-mean normals of which the two half-normals are their absolute values[2]. The regressors are assumed independent in probability from the error term. As was shown in the previous section, this model can describe various economic (and even non-economic) phenomena.

Stochastic frontier models are usually reparametrized, and in our case the most convenient reparametrization for the theoretical derivations proved to be the following:

$$\theta_1 \equiv \frac{\sigma_1}{\sigma_u},\ \theta_2 \equiv \frac{\sigma_2}{\sigma_u},\ s \equiv \sqrt{\sigma_u^2 + \sigma_1^2 + \sigma_2^2} = \sigma_u\sqrt{1+\theta_1^2+\theta_2^2}$$

while we use for compactness the following shorthands:

$$\omega_1 \equiv \frac{s\sqrt{1+\theta_2^2}}{\theta_1},\ \omega_2 \equiv \frac{s\sqrt{1+\theta_1^2}}{\theta_2},\ \lambda_1 \equiv \frac{\theta_2}{\theta_1}\sqrt{1+\theta_1^2+\theta_2^2},\ \lambda_2 \equiv \frac{\theta_1}{\theta_2}\sqrt{1+\theta_1^2+\theta_2^2}$$

With this notational conventions, the density of $\varepsilon_i$ was found to be

$$f_\varepsilon\left(\varepsilon_i\right) = \frac{2}{s}\phi\left(\varepsilon_i/s\right)\left[G_1\left(\varepsilon_i; 0, \omega_1, -\lambda_1\right) - G_2\left(\varepsilon_i; 0, \omega_2, \lambda_2\right)\right] \qquad [2]$$

[2] A word on notation related to the error terms may be in order: our notation differs from the one used by previous scholars, but it was chosen to satisfy many intuitive mnemonics. Firstly, the letters used to express the composite error term are placed in the order they have in the alphabet. Secondly, the benchmark model (with only a zero mean symmetric error) is placed first, and then is augmented by the terms that reflect the special nature of the 2TSF model (i.e. "base plus extension"). So $X\beta + u$ should be the benchmark model, and therefore $u$ represents the symmetric disturbance. Thirdly, to determine the order in which the one-sided error components appear, we used the fact that we usually think of the signs as "plus-minus" rather than as "minus-plus". So the positive error component appears first ($v$), the negative error component second ($w$). Fourthly, we associated the number 1 with the one-sided error that appears first (hence the $v$-component), and the number 2 with the one-sided error that appears second (the $w$-component), using the order in which the two numbers appear in the naturals.

where $\phi(\cdot)$ is the standard normal density while $G(z;\, location,\, scale,\, skew)$ is the cumulative distribution function (cdf) of a univariate skew-normal random variable. For compactness we will denote them simply $G_{1i}$ and $G_{2i}$, with general form

$$G_i \equiv G(\varepsilon_i;\, \xi,\, \omega,\, \lambda) = \int_{-\infty}^{\varepsilon_i} \frac{2}{\omega}\phi\left(\frac{t-\xi}{\omega}\right)\Phi\left(\lambda\frac{t-\xi}{\omega}\right)dt \qquad [3]$$

where $\Phi(\cdot)$ is the standard normal cdf. We will denote the corresponding densities by $g_{1i}$ and $g_{2i}$. The cdf of a skew normal with zero location parameter, as is our case, can be expressed as[3] $G(\varepsilon_i;\, 0,\, \omega,\, \lambda) = \Phi(\varepsilon_i/\omega) - 2T(\varepsilon_i/\omega;\, \lambda)$, where $T(\cdot\,;\cdot)$ is Owen's *T*-function defined, studied and tabulated in Owen (1956). Using this expression and the properties of the functions involved we can show the following:

When $\sigma_1^2 \to 0$ or $\sigma_2^2 \to 0$ we obtain a skew-normal density for the error term, which is the error density in the basic one-tier SF model ("production" or "cost" frontier respectively). If both variances go to zero, we obtain the zero-mean normal density. So the specification nests both the one-tier SF model, and the basic symmetric normal specification. Also, if $\sigma_1^2 = \sigma_2^2 = \sigma^2$ the density becomes symmetric around zero, exhibiting positive excess kurtosis compared to the normal distribution, and can be compactly written as

$$f_\varepsilon\left(\varepsilon_i\big|_{\sigma_1=\sigma_2=\sigma}\right) = \frac{8}{s}\phi(\varepsilon_i/s)T(\varepsilon_i/\omega_s\,;\,\lambda_s), \quad \omega_s = s\sqrt{1+\sigma_u^2/\sigma^2}, \quad \lambda_s = s/\sigma_u \qquad [4]$$

The existing exponential specification exhibits also excess kurtosis when it is symmetric.

### 3. Measuring the costs of inefficiency

The added value of stochastic frontier analysis lies in the measurement of the effects of inefficiency, with the 2TSF model going one step further and considering separately the two

[3] See Azzalini (1985).

sides of an economic transaction. Therefore, the actual measures of inefficiency used embody the core contribution of the approach. We examine two functional specifications, and for each we provide three sets of measures: sample-average measures, transaction-specific measures, and individual-specific measures. To focus ideas in this section we will treat the model as representing labor market transactions. The one-sided error terms are interpreted as reflecting the effects of imperfect/incomplete information, in the usual "search and match" context.

The estimation procedure itself will provide sample-average measures of the effects of informational inefficiencies on observed transactions. Jondrow et al. (1982) have introduced a method to derive also transaction-specific estimates of these effects, an approach which was extended for the 2TSF model by Kumbhakar and Parmeter (2009 and 2010), where the reader can find detailed discussions on the subject. This method requires that we calculate the distribution of each one-sided error term conditional on the composite error, then calculate the relevant conditional expected values (using the residuals of the estimation), and then use these expected values as estimates for the values of the one sided terms for each transaction. Under the half-normal specification we have obtained the following densities:

For the one-sided error $v$ (representing here the firm-buyer side of the transaction), the density of its distribution conditional on $\varepsilon$ is

$$f_{v|\varepsilon}\left(v_i\left|\varepsilon_i\right.\right)=\left(G_{1i}-G_{2i}\right)^{-1}\frac{2}{\omega_v}\phi\left(\frac{v_i}{\omega_v}-\frac{\varepsilon_i}{\omega_1}\right)\Phi\left(\lambda_1\frac{\left(v_i-\varepsilon_i\right)}{\omega_1}\right) \qquad [5]$$

where $\omega_v\equiv\frac{\sigma_1 s_2}{s}$, $s_2\equiv\sqrt{\sigma_2^2+\sigma_u^2}$ .

while for the one-sided error term *w* (representing the employee-seller side of the transaction), we have

$$f_{w|\varepsilon}\left(w_i\left|\varepsilon_i\right.\right)=\left(G_{1i}-G_{2i}\right)^{-1}\frac{2}{\omega_w}\phi\left(\frac{w_i}{\omega_w}+\frac{\varepsilon_i}{\omega_2}\right)\Phi\left(\lambda_2\frac{\left(w_i+\varepsilon_i\right)}{\omega_2}\right) \qquad [6]$$

where $\omega_w\equiv\frac{\sigma_2 s_1}{s}$, $s_1\equiv\sqrt{\sigma_1^2+\sigma_u^2}$ .

### *3.1. Levels Specification*

If the regression equation is specified in levels, and the error components enter additively, the estimated means of the two one-sided errors, $\hat{\mu}_v = \left(\sqrt{2/\pi}\right)\hat{\sigma}_1$ and $-\hat{\mu}_w = -\left(\sqrt{2/\pi}\right)\hat{\sigma}_2$, provide estimates of the sample-average deviations from the full-information wage, upwards or downwards respectively. Percentage deviations are obtained by dividing them by the estimated average full-information wage, which is $\overline{y} - \hat{\mu}_v + \hat{\mu}_w$ , where $\overline{y}$ is the sample average of the dependent variable. Summing the two (which equals the estimated mean of the composite error term), gives the net effect.

Transaction-specific measures are given by $E\left(v_i \middle| \varepsilon_i\right)$ (how much the full-information wage on transaction *i* was upwardly affected by the firm's (buyer) informational deficiencies) and $-E\left(w_i \middle| \varepsilon_i\right)$ (how much the full-information wage was downwardly affected by the employee-seller's informational deficiencies). The net effect on transaction *i* is their algebraic sum. In the half-normal specification, these conditional expected values are

$$E\left(v_i \middle| \varepsilon_i\right) = s^2\psi_{1i} + \sigma_1^2\left(\frac{1}{s^2}\varepsilon_i - \psi_i\right) \qquad [7]$$

$$E\left(w_i \middle| \varepsilon_i\right) = s^2\psi_{2i} - \sigma_2^2\left(\frac{1}{s^2}\varepsilon_i - \psi_i\right) \qquad [8]$$

where

$$\psi_{1i} \equiv \frac{g_{1i}}{G_{1i} - G_{2i}}, \quad \psi_{2i} \equiv \frac{g_{2i}}{G_{1i} - G_{2i}}, \quad \psi_i = \psi_{1i} - \psi_{2i} = \frac{g_{1i} - g_{2i}}{G_{1i} - G_{2i}}.$$

Kumbhakar & Parmeter (2009, 2010) introduced in addition relative *individual* measures of the costs of inefficiency, defined as percentage distances from *each-side's optimum*. This means that each is measured against a different benchmark, and so they are only conditionally comparable. For the firm-buyer side we measure (one minus) "best relative to actual" (where "best" here is "lowest cost", i.e. the reservation wage of the employee),

while for the employee-seller side we measure (one minus) "actual relative to best" (with "best" here meaning the highest wage that the firm would accept to pay).

For the specification in levels, the relative individual cost of informational inefficiency of a firm-buyer involved in transaction *i* is $\frac{E\left(v_i \middle| \varepsilon_i\right)}{y_i}$ , while the corresponding measure for the employee-seller involved in the same transaction is $\frac{E\left(w_i \middle| \varepsilon_i\right)}{y_i + E\left(w_i \middle| \varepsilon_i\right)}$ .

***3.2. Log Specification.***

If the dependent variable appears in its logarithm, while the error components enter again additively (this case covers both the usual semi-log and log-log regression specifications), then the sample-average *gross* mark-up (mark-down) effect of imperfections can be estimated by $\hat{E}\left(e^{v}\right)$ and by $\hat{E}\left(e^{-w}\right)$ correspondingly, while the combined net effect is multiplicative: $\hat{E}\left(e^{v}\right)\hat{E}\left(e^{-w}\right)-1$ . These unconditional expected values can be obtained from the moment generating function of the half normal random variables involved:

$$E\left(e^{v}\right) = 2\exp\left\{\frac{1}{2}\sigma_1^2\right\}\Phi\left(\sigma_1\right) = \sqrt{\frac{2}{\pi}}\frac{\Phi\left(\sigma_1\right)}{\phi\left(\sigma_1\right)} \qquad [9]$$

$$E\left(e^{-w}\right) = 2\exp\left\{\frac{1}{2}\sigma_2^2\right\}\Phi\left(-\sigma_2\right) = \sqrt{\frac{2}{\pi}}\frac{\Phi\left(-\sigma_2\right)}{\phi\left(\sigma_2\right)} \qquad [10]$$

Estimates are obtained by using $\hat{\sigma}_1$ , $\hat{\sigma}_2$ .

Transaction-specific effects are analogously given by $\hat{E}\left(e^{v_i} \middle| \varepsilon_i\right)$ (buyer's side) and by $\hat{E}\left(e^{-w_i} \middle| \varepsilon_i\right)$ (seller's side), and the net mark-up/mark-down effect by $\hat{E}\left(e^{v_i} \middle| \varepsilon_i\right)\hat{E}\left(e^{-w_i} \middle| \varepsilon_i\right)-1$.

Turning to the relative individual measures of the costs of inefficiency in each transaction *i*, they can be estimated by $1-\hat{E}\left(e^{-v_i} \middle| \varepsilon_i\right)$ for the firm-buyer side, and by $1-\hat{E}\left(e^{-w_i} \middle| \varepsilon_i\right)$ for the employee-seller side.

The three conditional expected values needed for the above transaction-specific and individual-specific measures are:

$$E\left(e^{-v}\middle|\varepsilon_i\right)=2\left(G_{1i}-G_{2i}\right)^{-1}\exp\left\{\frac{1}{2}\omega_v^2-\frac{\omega_v}{\omega_1}\varepsilon_i\right\}\left[\Phi\left(-\frac{\varepsilon_i+\sigma_1^2}{\omega_2}\right)-B\left(-\frac{\varepsilon_i+\sigma_1^2}{\omega_2},\,\omega_v-\frac{\varepsilon_i}{\omega_1};\,\rho^*\right)\right] \quad [11]$$

$$E\left(e^{y}\middle|\varepsilon_i\right)=2\left(G_{1i}-G_{2i}\right)^{-1}\exp\left\{\frac{1}{2}\omega_v^2+\frac{\omega_v}{\omega_1}\varepsilon_i\right\}\left[\Phi\left(-\frac{\varepsilon_i-\sigma_1^2}{\omega_2}\right)-B\left(-\frac{\varepsilon_i-\sigma_1^2}{\omega_2},\,-\left(\omega_v+\frac{\varepsilon_i}{\omega_1}\right);\,\rho^*\right)\right] \quad [12]$$

$$E\left(e^{-w}\middle|\varepsilon_i\right)=2\left(G_{1i}-G_{2i}\right)^{-1}\exp\left\{\frac{1}{2}\omega_w^2+\frac{\omega_w}{\omega_2}\varepsilon_i\right\}\left[\Phi\left(\frac{\varepsilon_i-\sigma_2^2}{\omega_1}\right)-B\left(\frac{\varepsilon_i-\sigma_2^2}{\omega_1},\,\omega_w+\frac{\varepsilon_i}{\omega_2};\,\rho^*\right)\right] \quad [13]$$

where $B\left(\cdot\,,\cdot\,;\,\rho\right)$ is the cumulative distribution function of a bi-variate standard normal with correlation coefficient $\rho$, while $\rho^*=\dfrac{-\sigma_1\sigma_2}{s_1 s_2}$ in all cases.

### 4. Model selection and maximum likelihood estimation

To validate statistically the use of the one-tier stochastic frontier model, it is natural to perform an ordinary least-squares estimation and then test the residuals for departures from normality, or perhaps more appropriately as argued by Schmidt and Lin (1984), to test separately for the existence of skewness. Godfrey and Orme (1991) and Bera and Premaratne (2001) have developed separate skewness tests that remain robust under non-normality. For the 2TSF model, due to its two-sided nature, absence of skewness should be followed by a test for excess kurtosis given zero skewness, since as we mentioned in Section 2, if $\sigma_1^2=\sigma_2^2=\sigma^2$ the two-tier specification will possess a density symmetric around zero, but will exhibit excess kurtosis. Finally, since the 2TSF nests both the one-tier and the symmetric normal specification, a Likelihood Ratio test could also be used (which is an advantage over the exponential specification).

Turning to matters of maximum likelihood estimation, we have proved that the log-likelihood is concave in the parameters, and so that, under the standard regularity conditions

the ML estimator will be consistent and asymptotically normal. The derivatives of the log-likelihood from an i.i.d sample are, in matrix notation where convenient,

$$\frac{\partial \tilde{L}}{\partial \boldsymbol{\beta}} = \frac{1}{s^2}\left(\mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\boldsymbol{\beta}\right) - \mathbf{X}'\boldsymbol{\psi} \qquad [14]$$

$$\frac{\partial \tilde{L}}{\partial s} = \frac{1}{s}\left(\frac{1}{s^2}\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} - n - \boldsymbol{\varepsilon}'\boldsymbol{\psi}\right) = \frac{1}{s}\left(\frac{1}{s^2}\sum_{i=1}^{n}\varepsilon_i^2 - n - \sum_{i=1}^{n}\varepsilon_i\psi_i\right) \qquad [15]$$

$$\frac{\partial \tilde{L}}{\partial \theta_1} = \frac{\theta_1}{1+\theta_1^2}\sum_{i=1}^{n}\left\{\sqrt{\frac{2}{\pi}}\frac{1}{\lambda_2}\phi\left(\frac{\sqrt{\theta_1^2+\theta_2^2}}{s}\varepsilon_i\right)\left(G_{1i}-G_{2i}\right)^{-1} + \left(\frac{1+\theta_1^2}{\theta_1^2}\psi_{1i} + \psi_{2i}\right)\varepsilon_i\right\} \qquad [16]$$

$$\frac{\partial \tilde{L}}{\partial \theta_2} = \frac{\theta_2}{1+\theta_2^2}\sum_{i=1}^{n}\left\{\sqrt{\frac{2}{\pi}}\frac{1}{\lambda_1}\phi\left(\frac{\sqrt{\theta_1^2+\theta_2^2}}{s}\varepsilon_i\right)\left(G_{1i}-G_{2i}\right)^{-1} - \left(\psi_{1i} + \frac{1+\theta_2^2}{\theta_2^2}\psi_{2i}\right)\varepsilon_i\right\} \qquad [17]$$

where we have kept $\varepsilon_i = y_i - \mathbf{x}_i'\boldsymbol{\beta}$ for compactness, and where $\boldsymbol{\psi}$ is a $n \times 1$ column vector containing the $\psi_i$ terms.

Closing, we mention that for purposes of empirical estimation, the density $f_\varepsilon(\varepsilon_i)$ and the derivatives of the log-likelihood can be easily transformed and coded as functions of the three base scale parameters $\sigma_u$, $\sigma_1$, $\sigma_2$ instead of their reparametrized counterparts. Reverting back to the fundamental parameters has the advantage of providing directly estimates of the asymptotic standard errors. Also, one can show that the skew-normal cdf has the alternative representation

$$G(\varepsilon_i;\, \xi,\, \omega,\, \lambda) = 2B\left(\frac{\varepsilon_i - \xi}{\omega},\, 0\,;\, \rho = \frac{-\lambda}{\sqrt{1+\lambda^2}}\right) \qquad [18]$$

i.e. it can be expressed as a function of the bivariate standard normal cdf. This expression makes $f_\varepsilon(\varepsilon_i)$ a function only of the standard normal density and the standard normal bivariate cdf, the second being more widely implemented in software packages compared to Owen's *T*-function.

**5. Conclusions and further research**

The Two-tier Stochastic Frontier model has recently seen an increase in the number of studies in which it is used, and more importantly, in the areas of economic (and even non-economic) activity in which it is applied, bearing out the predictions of the scholars that have implemented it during its first phase of existence. In this paper, we have presented a new stochastic specification for the model, the half-normal, where the two one-sided error terms are assumed to follow half-normal distributions. In an empirical study, not presented here, we have compared results from the existing exponential specification with the half normal one. For the specific data set the composite error distribution appeared symmetric with excess kurtosis. The two specifications were almost identical as regards the estimated regressor coefficients and the maximized value of the likelihood. But they produced different estimates for the scale parameters of the one-sided error-terms, and hence different estimates for the various measures of inefficiency presented previously. We conjecture that the presence of skewness will intensify the distance between the two specifications.

Therefore, a fruitful area of further research would be to develop procedures for the evaluation of, and selection between, the two specifications, possibly suited to the specific stochastic frontier framework, apart from the available general methods for non-nested model selection. But the flexibility and applicability of the Two-tier stochastic frontier model through additional stochastic specifications can be further enriched, and for good reason: one direction is to obtain hybrid setups, like an exponential/half normal or a half-normal/exponential specification. As mentioned earlier, in Tsionas (2012) the exponential specification was rejected for one of the two one-sided error terms but not for the other and this is an indication that inefficiencies may follow different distributions in each side of an economic transaction.

Another related direction of research is to obtain closed-form specifications for the one-sided error terms that nest the existing ones. For the half normal case, this would be the truncated normal specification. Such a specification would permit to *test* whether the modes

of the distributions of the one-sided error terms are non-zero. Both the exponential and the half-normal densities have their mode at zero, and, as it has been noted early on by Stevenson (1980) for the one-tier stochastic frontier model, this imposes a restrictive a priori assumption regarding the likelihood of inefficiencies characterizing market participants.

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